

Improved physics-informed neural networks for the reinterpreted discrete fracture model

Chao Wang^{a,*}, Hui Guo^{a,*,1}, Xia Yan^{b,2}, Zhang-Lei Shi^a, Yang Yang^{c,*,3}

^a College of Science, China University of Petroleum (East China), Qingdao, Shandong Province, 266580, China

^b School of Petroleum Engineering, China University of Petroleum (East China), Qingdao, Shandong Province, 266580, China

^c Department of Mathematical Sciences, Michigan Technological University, Houghton, MI 49931, USA

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ABSTRACT

This paper is the first attempt to apply improved-physics-informed neural networks (I-PINNs) to simulate fluid flow in fractured porous media based on the reinterpreted discrete fracture model (RDFM). The RDFM, first introduced by Xu and Yang, is a hybrid-dimensional model where Dirac-delta functions are used to characterize fractures and superposed with the permeability tensor. In this paper, we apply the physical information neural networks (PINNs) to RDFM. Different from the traditional PINNs where the PDE residual was used as the loss function, we adopt the finite element discretization of RDFM to build the loss function, avoiding the large gradient problem and difficulties in automatic differentiation. This new method is named as the improved PINNs (I-PINNs). Moreover, we combine the RDFM with incompressible miscible displacement in porous media. The bound-preserving technique of the I-PINNs is proposed and applied to the coupled system mentioned above, keeping the numerical concentration to be between 0 and 1. It is worth noting that one of the advantages of I-PINNs compared to PINNs is that it can better capture the pressure gradient at the fractures. Compared with traditional finite element methods for flow equations, I-PINNs do not request the inversion of the stiffness matrix. In addition, different from the traditional bound-preserving technique for contaminant transportation, I-PINNs preserve the physical bounds without taking a limited time step. Several numerical experiments are given to verify the feasibility and accuracy of the I-PINNs.

1. Introduction

In recent years, the study of fluid flow in fractured porous media has attracted a lot of attention due to the popularity of hydraulic fracturing technology in unconventional reservoir development. However, a variety of factors such as the vast difference in permeability between fractures and matrix, the complexity of the fracture network's geometric shapes, among others, present significant difficulties and challenges in modeling and simulating the problem. Hence, it is essential to accurately establish a mathematical model of fluid flow in fractured porous media as well as an efficient numerical method.

* Corresponding authors.

E-mail addresses: sdugh@163.com (H. Guo), yyang7@mtu.edu (Y. Yang).

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For these reasons, in recent decades researchers have developed a number of models depending on the treatment of the fractures, which can be categorized as continuum models and discrete fracture-matrix models. The dual porosity model [1–4] is a typical continuum media model, which treats the matrix and fracture network as a continuous system with different permeabilities, governed by Darcy's Law. However, it generates certain errors when simulating disconnected fractured media. Due to the aforementioned reasons, researchers have developed a range of discrete fracture-matrix models to accurately describe individual fracture, including: the discrete fracture model (DFM) [5–10], embedded discrete fracture model (EDFM) [11–14], extended finite element DFM (XFEM-DFM) [15–17], and so on.

The main idea of the DFM is to view the fractures as $s-1$ dimensional entries on the interface of s -dimensional matrix cells, with the thickness of fractures being the dimensional homogeneity factor. Since the fractures must be located at the edges of the matrix elements, it is difficult to generate high-quality meshes when the fracture network is very complex. Therefore, embedded DFM (EDFM) has emerged as an alternative solution. It successfully avoids the mesh limitation and is widely used in real reservoir problems. Another class of discrete fracture-matrix models is XFEM-DFM. It imposes interface conditions across fractures by enriching the function space in the elements that intersect the fractures.

In 2020, Xu and Yang [18] proposed a reinterpreted discrete fracture model (RDFM), which uses the Dirac- δ function to characterize the permeability tensor of fractures and provides a new hybrid-dimensional representation of the permeability tensor. Its advantage lies in avoiding the mesh restrictions imposed by traditional DFM. Additionally, fractures and matrix are coupled in one system, and the computational cost does not increase with the increasing number of fractures. Building on this work, Guo et al. [19] proposed conservative numerical methods for the RDFM and extended them to the contaminant transportation. Xu et al. [20,21] and Fu et al. [22,23] used RDFM to simulate flow in porous media containing both fractures and barriers. After that, Wu et al. [24] extended this novel model to Darcy-Forchheimer flow in fractured porous media.

With the explosive growth of computing resources in the past decade, machine learning methods have undergone revolutionary development, such as the deep neural networks (DNNs) [25–27], the convolutional neural networks (CNNs) [28–30] and the recurrent neural networks (RNNs) [31–33], etc. In recent years, a novel DNN framework has been developed, called Physics-informed neural networks (PINNs) [34]. This approach was first proposed to solve the forward and inverse problems described by the combination of some data and partial differential equations (PDEs) [35]. It has been utilized to various fluid mechanics [36], elasticity [37] and seepage mechanics problems [38]. Some attractive features of the PINNs are summarized as follows: (a.) Compared with traditional numerical methods, PINN belongs to the category of meshless methods, so it can easily deal with complex mobile domain problems; (b.) The employment of the PINNs provides closed-form and differentiable solutions which can be easily used in any subsequent calculation; (c.) Based on embedded physical laws (basic PDE), PINN can be trained with a small amount of measured data; (d.) The code is easy to implement by using existing open-source deep learning algorithms.

Though PINN is a powerful tool in solving PDEs, it cannot yield accurate numerical approximations for RDFM, and we will use numerical experiments to demonstrate this. In this paper, we propose the improved-physics-informed neural networks (I-PINNs) framework suitable for RDFM and extend them to contaminant transportation. We have made some significant breakthroughs in the above work: (a.) The I-PINNs framework proposed in this paper for solving single-phase flow in fractured porous media takes the coordinates of collection points as input and the pressure field as output, establishing a map from coordinates to pressure and possessing good surrogate model characteristics [38]. (b.) When calculating the single-phase flow problem in porous media with fractures, the PINNs' automatic differentiation technique is inevitably needed to calculate the pressure gradient at the fractures, and such technique may fail to work [39]. In the I-PINNs' framework, the finite element discretization was used to represent the pressure gradient, eliminating the difficulties in automatic differentiation. (c.) We use a finite element discretization scheme to construct an approximation of the PDE residual loss function, avoiding the penalty terms of adjusting the loss from the boundary loss and the control equation in the traditional PINNs, and improving the efficiency [40]. (d.) Physically, the concentration in the contaminant transportation equation is between 0 and 1. Therefore, we propose bound-preserving I-PINNs in our paper. Compared with the traditional bound-preserving techniques, e.g. [41], our technique does not require a limited the time step size dt , which greatly increases the computational efficiency and reduces the computational cost [42].

The rest of this article is as follows. Section 2 reviews the RDFM for single-phase flow in fractured porous media. Section 3 describes the I-PINNs framework and applies it to RDFM. In Section 4, we use several numerical experiments to demonstrate the effectiveness and accuracy of the proposed technique. The application in contaminant transportation is shown in Section 5. Finally, we make some concluding statements in Section 6.

2. Problem statement

2.1. Equi-dimensional model for single-phase flow in fractured media

We consider the steady-state single-phase flow in heterogeneous porous media, which has the domain $\Omega \subseteq R^N$ and boundary $\partial\Omega$. The governing equation of this problem can be expressed as

$$-\nabla \cdot (\mathbf{K} \nabla p) = f, \quad x \in \Omega, \quad (2.1)$$

where $\mathbf{K}$ is the permeability tensor of the porous media and f is the source term. For fractured porous media, $\mathbf{K}$ can be divided into

$$\mathbf{K} = \begin{cases} \mathbf{K}_m & x \in \Omega_m \\ \mathbf{K}_f & x \in \Omega_f, \end{cases} \quad (2.2)$$

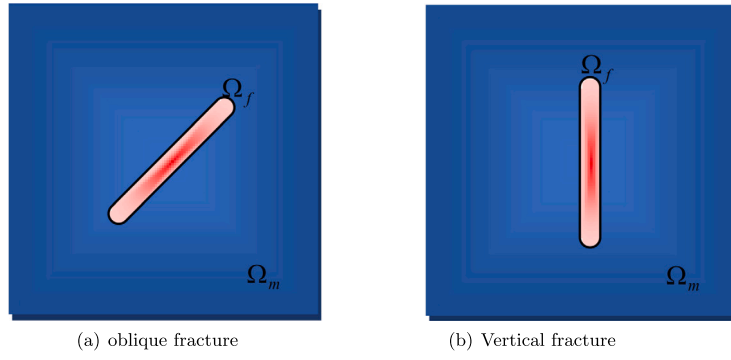


Fig. 1. Fractured structure.

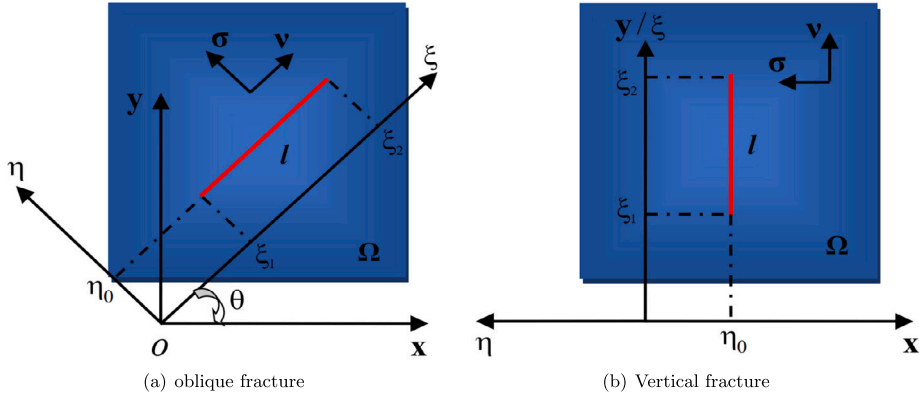


Fig. 2. Fractured structure and the corresponding coordinate systems.

in which Ω_m and Ω_f represent the domains of porous matrix and fractures, respectively. For the convenience of observation, Fig. 1 shows the model after the thickness of the fracture is magnified.

The Dirichlet and Neumann boundary conditions are as follows

$$p = p_D, \quad \text{on } \Gamma_D, \quad (2.3)$$

$$(\mathbf{K} \nabla p) \cdot \mathbf{n} = q_N, \quad \text{on } \Gamma_N = \partial\Omega \setminus \Gamma_D, \quad (2.4)$$

where $\mathbf{n}$ is the unit outer normal vector of $\partial\Omega$.

It is worth noting that (2.1)–(2.4) can be summarized as the equi-dimensional model for single-phase flow in fractured media, and $\mathbf{K}$ in (2.2) is the equi-dimensional representation of permeability tensor.

2.2. Hybrid-dimensional model for single-phase flow in fractured media

In this section, we review the RDFM proposed by Xu and Yang [18] to express the permeability $\mathbf{K}$ of fractured porous media as

$$\mathbf{K} = \mathbf{K}_m + \epsilon k_f \delta(\eta - \eta_0) \mathbb{1}(\xi_1 \leq \xi \leq \xi_2) \mathbf{v} \mathbf{v}^T, \quad (2.5)$$

where $\mathbf{K}_m$, ϵ , k_f are the permeability tensor of matrix, the thickness of the fracture, and the tangential permeability of the fracture, respectively. $\mathbb{1}(\cdot)$ the indicator function defined as $\mathbb{1}(\text{expr})$ equals 1 if expr is true while equals 0 otherwise. $\delta(\eta - \eta_0) \mathbb{1}(\xi_1 \leq \xi \leq \xi_2)$ indicates the location of the fracture. We give the global xOy and local coordinates systems $\eta O \xi$ in Fig. 2, where $\mathbf{v}$ and $\boldsymbol{\sigma}$ represent the tangential and normal unit vector of fracture l , respectively. The permeability $\mathbf{K}$ in the global coordinate system xOy can be expressed as

$$\mathbf{K} = \mathbf{K}_m + \epsilon k_f \delta(-\sin(\theta)x + \cos(\theta)y - \eta_0) \mathbb{1}(\xi_1 \leq \cos(\theta)x + \sin(\theta)y \leq \xi_2) \mathbf{v} \mathbf{v}^T. \quad (2.6)$$

For the system containing multiple fractures, we consider the following equation

$$\mathbf{K} = \mathbf{K}_m + \sum_{i=1}^L \epsilon_i k_{f_i} \delta_i(\cdot) \mathbb{1}_i(\cdot) \mathbf{v}_i \mathbf{v}_i^T, \quad (2.7)$$

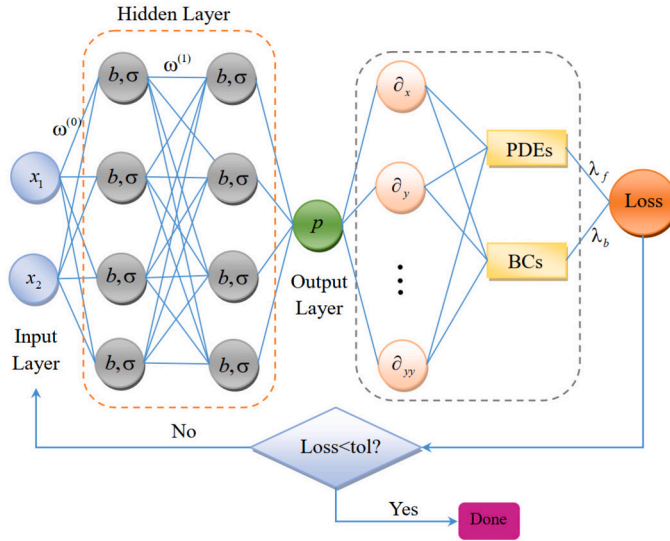


Fig. 3. Schematic diagram of physics-informed neural networks (PINNs).

in which L denotes the number of fractures. $\delta(\cdot)\mathbb{1}(\cdot)$ is the abbreviated form of the corresponding expressions in (2.5) and (2.6). ϵ_i, k_{fi}, v_i are the thickness, the tangential permeability and the tangential unit vector of the i th fracture, respectively. Finally, we summarize the hybrid-dimensional model of single-phase flow in fractured porous media as follows

$$\begin{aligned}
 -\nabla \cdot (\mathbf{K} \nabla p) &= f, & x \in \Omega, \\
 \mathbf{K} &= \mathbf{K}_m + \sum_{i=1}^L \epsilon_i k_{fi} \delta_i(\cdot) \mathbb{1}_i(\cdot) v_i v_i^T, & x \in \Omega, \\
 p &= p_D, & \text{on } \Gamma_D, \\
 (\mathbf{K} \nabla p) \cdot \mathbf{n} &= q_N, & \text{on } \Gamma_N.
 \end{aligned} \tag{2.8}$$

3. Methodology

3.1. The fully-connected neural network

We start with a brief review of traditional PINNs. Fig. 3 illustrates the architecture of a multi-layer fully-connected operational neural network containing input layers corresponding to inputs x_1 and x_2 , a hidden layer of several neurons, and an output layer corresponding to the output p . The trial solution of the problem can be approximated by trainable parameters such as the weights ω and biases b of the neural structure:

$$\begin{aligned}
 p &= \sum_{j=1}^{M_2} \omega_j^{(2)} \sigma(z_j^{(1)}) + b^{(2)}, \\
 z_j^{(1)} &= \sum_{i=1}^{M_1} \omega_{ji}^{(1)} \sigma(z_i^{(0)}) + b_j^{(1)}, \\
 z_i^{(0)} &= \omega_{i1}^{(0)} x_1 + \omega_{i2}^{(0)} x_2 + b_i^{(0)},
 \end{aligned}$$

where M_1 and M_2 are the number of the neuron units in the first and second hidden-layers, respectively. $\omega_{ij}^{(0)}$ represents the weight from the input unit x_j to the i th neuron of the first hidden-layer, $\omega_{ji}^{(1)}$ denotes the weight from the i th neuron of the first hidden-layer to the j th neuron of the second hidden-layer, $\omega_j^{(2)}$ stands for the weight from the j th neuron of the second hidden-layer to the output layer, $b_i^{(0)}$, $b_j^{(1)}$ and $b^{(2)}$ are the neuron biases of the first hidden-layer, second hidden-layer and the output layer, respectively. $\sigma(\cdot)$ expresses the nonlinear activation function. This can be extended to the case with more hidden layers.

3.2. Loss function for the PINNs

In the traditional PINNs, the trainable parameters ω and b of the network can be “learned” by minimizing a loss function of the following general form:

$$Loss = \lambda_f Loss_f + \lambda_b Loss_b, \quad (3.1)$$

where

$$Loss_f = \frac{1}{N_f} \sum_{i=1}^{N_f} |lhs_i - f_i|^2,$$

$$Loss_b = \frac{1}{N_b} \sum_{i=1}^{N_b} |p_i - \bar{p}_i|^2.$$

The loss function (3.1) measures how well the trial solution p satisfies the corresponding PDE and boundary conditions. The goal is to find a set of parameters ω and b such that the loss function (3.1) is as small as possible. According to the above equation, $Loss_f$ and $Loss_b$ are the losses of the PDE and the boundary conditions, N_f , N_b represent the numbers of nodes that satisfy the PDE and boundary conditions, respectively. λ_f and λ_b are the weights of the losses of the PDE and the boundary conditions, respectively. These weights, which can be defined by the researchers themselves or automatically adjusted by adaptive techniques, play a key role in improving the performance of PINNs. Thus, the original problem can be interpreted as an optimization problem, i.e., the network is trained by various gradient-based optimization techniques using the minimized loss function (3.1).

3.3. The PINNs for the hybrid-dimensional model for single-phase flow in fractured media

Based on the ideas in [43], Eq. (2.1) is transformed into the form of an energy function. Thus, the loss of the control equation in Eq. (3.1) can be rewritten as

$$Loss_f = \int_{\Omega} \left(\frac{1}{2} \mathbf{K} \nabla p \cdot \nabla p - f p \right) dx. \quad (3.2)$$

For convenience, the flow is driven by boundary conditions rather than source terms, that is, $f = 0$. Combining with Eq. (2.7), the above equation can be further expressed as

$$Loss_f = \int_{\Omega} \left(\frac{1}{2} (\mathbf{K}_m + \epsilon k_f \delta(\cdot) \mathbb{1}(\cdot) v v^T) \nabla p \cdot \nabla p \right) dx = \int_{\Omega} \frac{1}{2} \mathbf{K}_m \nabla p \cdot \nabla p dx + \int_{\Omega} \frac{1}{2} \epsilon k_f \delta(\cdot) \mathbb{1}(\cdot) v v^T \nabla p \cdot \nabla p dx. \quad (3.3)$$

According to Lemma 4.1 in [18], we rewrite the integral of the fracture term as the line integral along the fracture. Then, the transformed equation is as follows

$$Loss_f = \int_{\Omega} \frac{1}{2} \mathbf{K}_m \nabla p \cdot \nabla p dx + \sum_{i=1}^L \int_{l_i} \frac{1}{2} \epsilon_i k_{fi} \frac{\partial p}{\partial v_i} \frac{\partial p}{\partial v_i} ds. \quad (3.4)$$

Therefore, the problem is converted into estimating the minimum value of the above energy function.

3.4. Improved-physics-informed neural networks (I-PINNs)

Based on the idea of PINNs, we develop the I-PINNs framework to simulate the RDFM. In conventional PINNs, the outputs are substituted into the control equations and the residuals are computed by automatic differentiation (AD). AD is commonly used to train neural networks by calculating the derivative of a function at a point, rather than by analytically deriving an expression for the derivative of the function. We know that high gradients pose a serious challenge to the accuracy of AD, and it has not yet been shown in the literature that the continuity of fluxes between mesh cells of different properties can be accurately portrayed using AD.

In I-PINNs, we first discretize the governing equations (2.8) by FEM. Then the loss function is evaluated based on FEM. By doing so, the PDE residuals in PINNs are approximated by substituting the output of the neural network into the numerical schemes instead of the original PDEs, which avoids automatic differentiation and thus improves the accuracy.

In view of the foregoing thoughts, we first establish the variational form of the hybrid-dimensional model (2.8): to find $p \in H_D^1$ such that for all $v \in H_0^1$ we have

$$\int_{\Omega} (\mathbf{K}_m + \epsilon k_f \delta(\cdot) \mathbb{1}(\cdot) v v^T) \nabla p \cdot \nabla v dx dy = \int_{\Omega} f v dx dy + \int_{\Gamma_N} q_N v ds, \quad (3.5)$$

in which H_D^1 and H_0^1 denote the variational spaces $H_D^1 := \{v \in H^1(\Omega) : v|_{\Gamma_D} = p_D\}$ and $H_0^1 := \{v \in H^1(\Omega) : v|_{\Gamma_D} = 0\}$, respectively. The above equation can then be reduced to the following equivalent form

$$\int_{\Omega} \mathbf{K}_m \nabla p \cdot \nabla v dx dy + \int_{\Omega} \epsilon k_f \delta(\cdot) \mathbb{1}(\cdot) \frac{\partial p}{\partial v} \frac{\partial v}{\partial v} dx dy = \int_{\Omega} f v dx dy + \int_{\Gamma_N} q_N v ds. \quad (3.6)$$

By Lemma 4.1 in [18], we rewrite the integral of the fracture term as the line integral along the fracture. Then, the transformed equation is as follows

$$\int_{\Omega} \mathbf{K}_m \nabla p \cdot \nabla v dx dy + \int_l \epsilon k_f \frac{\partial p}{\partial v} \frac{\partial v}{\partial v} ds = \int_{\Omega} f v dx dy + \int_{\Gamma_N} q_N v ds. \quad (3.7)$$

It should be noted that the shape of the fracture can be a curve, and the thickness ϵ and tangential permeability k_f are scalars. For a fracture network composed of multiple fractures, the following formula is available

$$\int_{\Omega} \mathbf{K}_m \nabla p \cdot \nabla v dx dy + \sum_{i=1}^L \int_{l_i} \epsilon_i k_{fi} \frac{\partial p}{\partial v_i} \frac{\partial v}{\partial v_i} ds = \int_{\Omega} f v dx dy + \int_{\Gamma_N} q_N v ds, \quad (3.8)$$

in which L is the number of the fractures.

Then the finite element scheme of the RDFM is constructed by using the tensor product linear functions as the finite element space for rectangular meshes in the variational form. For triangular meshes, the traditional linear basis can be applied. Thus, we obtain the following linear system

$$\sum_{j=1}^N \left\{ \int_{\Omega} (\mathbf{K}_m \nabla \Psi_j) \cdot \nabla \Psi_j dx dy + \int_l \epsilon k_f \frac{\partial \Psi_j}{\partial v} \frac{\partial \Psi_j}{\partial v} ds \right\} p_j = \int_{\Omega} f \Psi_j dx dy + \int_{\Gamma_N} q_N \Psi_j ds, i = 1, 2, \dots, M. \quad (3.9)$$

In addition, the Dirichlet boundary condition is

$$p_j = p(\mathbf{x}_j), \quad \mathbf{x}_j \in \Gamma_D, \quad j = M + 1, \dots, N, \quad (3.10)$$

where Ψ_j are the basis, M is the number of non-Dirichlet vertices and N is the total number of mesh vertices.

Furthermore, the output of the NN is substituted into (3.9) to approximate the PDE residual which should be minimized. Therefore, mean absolute error (MAE) loss function for the NN is defined as

$$Loss_f = \frac{1}{M} \sum_{i=1}^M \left| \sum_{j=1}^N \left\{ \int_{\Omega} (\mathbf{K}_m \nabla \Psi_j) \cdot \nabla \Psi_j dx dy + \int_l \epsilon k_f \frac{\partial \Psi_j}{\partial v} \frac{\partial \Psi_j}{\partial v} ds \right\} p_j^* - \int_{\Omega} f \Psi_j dx dy - \int_{\Gamma_N} q_N \Psi_j ds \right|, \quad (3.11)$$

where p_j^* is the j th element of the NN output. The NN is trained by minimizing MAE to obtain the solution of p_j , $j = 1, 2, \dots, N$. Since the boundary conditions are imposed directly in (3.9), $Loss_b$ is not used to train the NN, and only $Loss_f$ is taken into account in the loss function, i.e. the Dirichlet boundary condition is applied by manually setting the boundary output to p_D . Finally, we illustrate the I-PINNs framework in Fig. 4.

4. Numerical experiments

In this section, we simulate several benchmark single-phase flow problems in fractured porous media to verify the accuracy and convergence of the I-PINNs. For simplicity, the flow is driven by boundary conditions instead of source term, i.e., $f = 0$. In addition, all the numerical experiments are built on the Pytorch deep learning environment with Relu activation function and torch.optim.Adam optimization method [38]. We ran all the examples on computers with i5-13500HX 2.50 GHz CPU, GTX 4060 GPU, 16 GB memory, 500 GB hard drive and a Windows 11 operating system. The performance of the I-PINNs was evaluated according to the following definition of absolute error:

$$L_{\text{error}} = |P_{\text{pred}} - P_{\text{ref}}|, \quad (4.1)$$

where P_{pred} and P_{ref} denote the numerical and reference solutions, respectively. P_{ref} is given by using the standard Galerkin finite element method on a fully resolved rectangular grid ($N_x = N_y = 1001$). If not otherwise stated, in all the numerical examples, we use 2 hidden layers and 200 neurons per layer.

Example 4.1 (Verification of accuracy). In the first example, we verify the accuracy of the proposed method by analyzing a Poisson problem with smooth heterogeneous conductivity. The computational region is a square plate of length 1 (shown in Fig. 5(a)), and the conductivity is $k(x) = \exp(-2x)$. The left and right boundary conditions are $p_D = 10, p_D = 0$, respectively, while the lower and upper boundaries are insulated.

We use the uniform rectangular meshes ($N_x = N_y = 40$) to discretize the entire computational domain (as shown in Fig. 5(b)). Fig. 6 clearly shows that FEM, PINN, and I-PINN can all simulate the problem very well. In addition, their results are consistent, verifying the accuracy of our proposed method.

Example 4.2 (Cross-shaped fractures). In this example, we consider a square structure with a simple cross-shaped fractures. The computational domain is $[0, 1] \times [0, 1]$ with cross-shaped fractures $[0.25, 0.75] \times [0.4995, 0.5005] \cup [0.4995, 0.5005] \times [0.25, 0.75]$ laying on it. The permeabilities of fractures and matrix are 10^4 and 1, respectively. The boundary conditions are shown in Fig. 7(a).

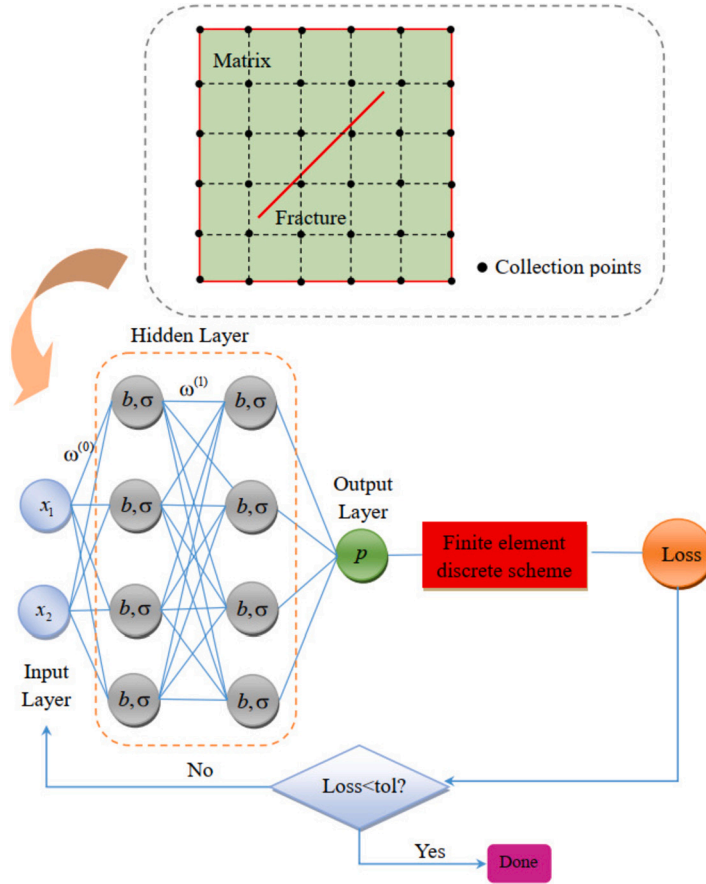


Fig. 4. Schematic diagram of I-PINNs.

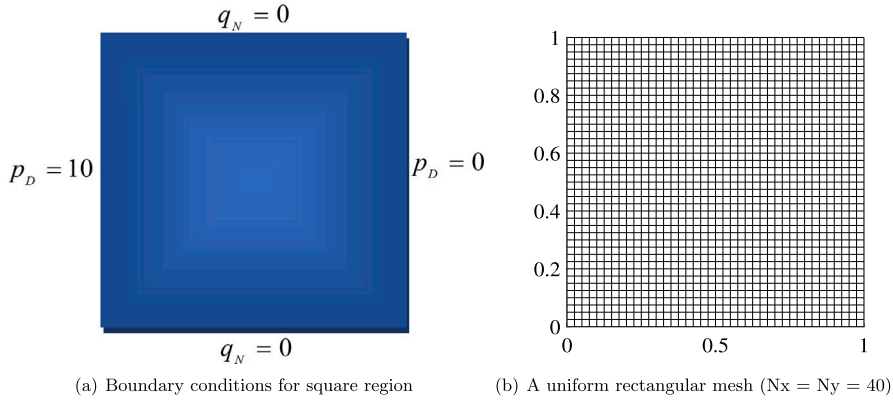


Fig. 5. Boundary conditions and mesh profiles.

We use the uniform rectangular meshes ($N_x = N_y = 30$) to discretize the entire computational domain (as shown in Fig. 7(b)). We train the network by minimizing the loss function (3.11) and iterating for 200000 steps. It can be seen from Fig. 8(a) that the numerical approximation of pressure in the computational domain agrees with the reference solution (Fig. 8(b)) given in [44]. Moreover, we plot the pressure slices along $y = 0.45$ and $x = 0.3$ in Fig. 9. We can see that the results for I-PINNs are consistent with the reference solution. The maximum errors between the I-PINNs and the reference solutions on the two line segments are 4.551×10^{-2} and 1.198×10^{-2} , respectively. It should be noted that we tried to modify the neural network structure parameters (such as the number of neurons and the number of layers) and found that these changes had little impact on the calculation results. Furthermore, the change of MAE training loss over all epochs is plotted in Fig. 10, showing that the lowest loss is around 7.60×10^{-4} .

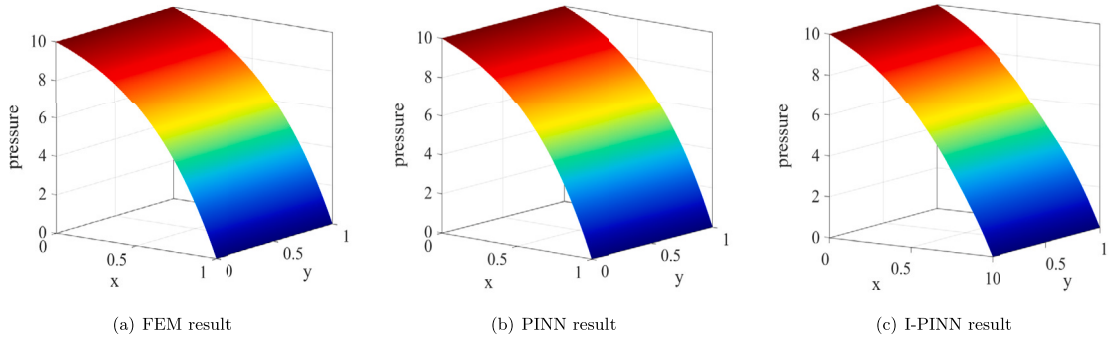


Fig. 6. Comparison of results for FEM, PINN, and I-PINNs.

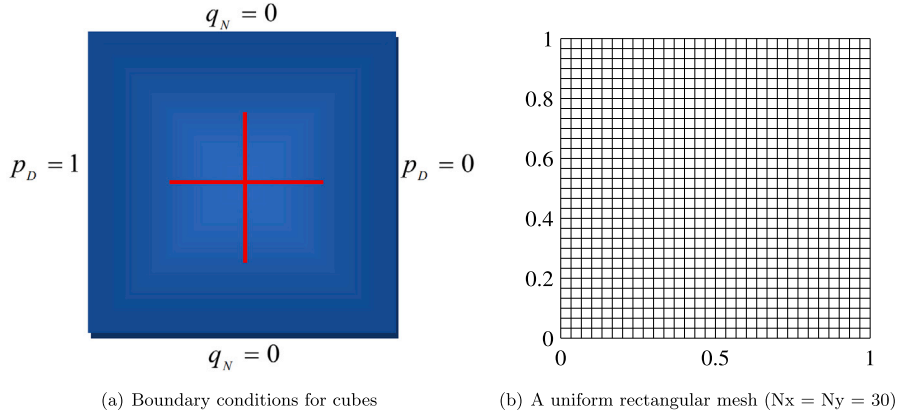


Fig. 7. Boundary conditions and mesh profiles.

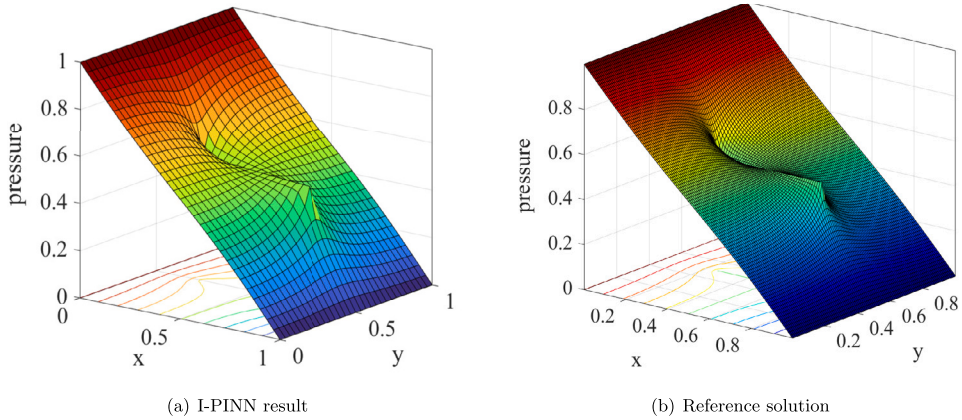


Fig. 8. Reference solution as well as I-PINNs result.

To validate the advantages of the novel I-PINNs framework proposed in this paper as well as the fracture model described using the Dirac- δ function. First, the traditional PINNs simulation of an equi-dimensional model of single-phase flow in fractured porous media is considered (see (2.1)–(2.4)). Keeping the computational domain as well as the boundary conditions consistent with the above problem, we use the same number of network layers and neurons as before, and iterate 300000 steps. The computational domain is discretized using 1681 uniformly distributed nodes, including 1640 matrix nodes and 41 fracture nodes. The distribution of nodes as well as the computational results are shown in Fig. 11(a), and Fig. 11(b), respectively. From Fig. 11(b), it can be seen that the effect the fracture can hardly be observed, which is due to the fact that conventional PINNs use discrete node information, and a single line does not clearly represent the fracture. To clearly depict the effect of the fractures, more nodes must be assigned along of the fracture (as shown in Fig. 11(c)), which increases the computational cost. In fact, we find in Fig. 11(d) that even with the increased number of

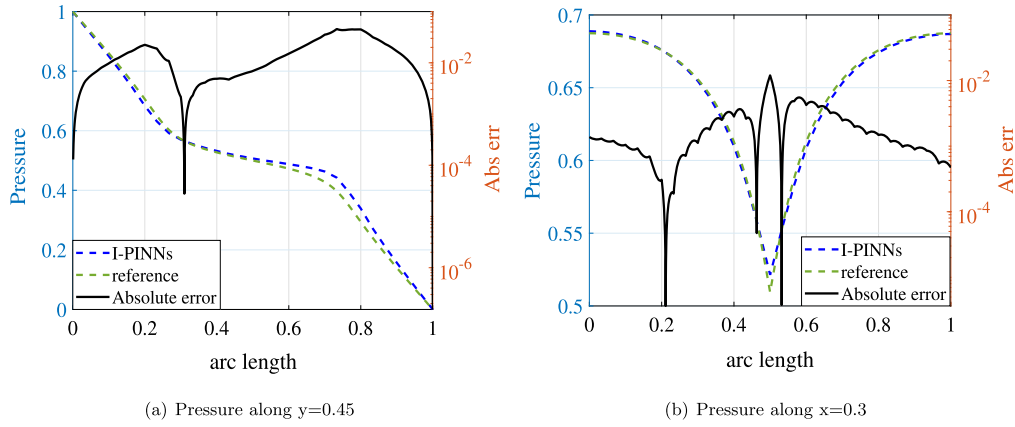


Fig. 9. Pressure profiles along different lines.

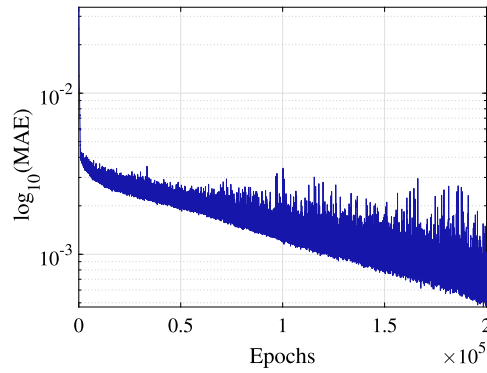


Fig. 10. The convergence of the MAE loss over epochs.

nodes, there is still no significant change in the pressure distribution. Further investment is needed to understand why PINNs failed to work for the equi-dimension model.

To address the above problems, we consider RDFM (see Eq. (2.8)). We simulate the problem using conventional PINNs, treated in a manner consistent with the problem above. Fig. 12(a) clearly shows that the model using the Dirac- δ function is able to describe the pressure distribution in the fracture, without considering the accuracy. This is due to the fact that although we still use nodal information, the fracture is clearly characterized in the hybrid-dimensional model. A comparison with Fig. 11(b) shows that methods like the PINNs, which consider discrete nodal information, have a great advantage in using Dirac- δ functions to describe the fracture. However, the difficulty in accurately describing large gradients due to the automatic differentiation technique used in PINNs and the need to set the values of the individual weights in the loss function make this result still less satisfactory. Therefore, we propose the I-PINNs based on the above work. We use 1600 rectangular cells, so that the number of mesh vertices is still 1681. The same optimization approach given above as well as the parallel techniques are used. The numerical pressure is shown in Fig. 12(b), which is more consistent with the reference solution. Finally, a comparison of pressure slice of the reference solution, the I-PINNs, and the PINNs with Dirac- δ function along $y = 0.5$ is shown in Fig. 13. We find that the results of the I-PINNs are significantly better than those of the conventional PINNs.

Example 4.3 (Curved fractures). In this example, we simulate a round closed fracture to analyze the feasibility and effectiveness of the I-PINNs frame for curved fractures. The computational domain is a unit square. The parameter equations of the fracture are given as

$$\begin{cases} x_1(t) = \frac{1}{2} + \frac{1}{4} \cos(t) \\ y_1(t) = \frac{1}{2} + \frac{1}{4} \sin(t), \end{cases} \quad 0 \leq t \leq 2\pi.$$

The permeabilities of porous media and fracture are set to be 1 and 10^4 , respectively. The thickness of fractures is 0.0025. We impose Dirichlet boundary conditions $p_D = 1$ and $p_D = 0$ on the left and right boundaries. The other boundaries are impermeable (as shown in Fig. 14(a)).

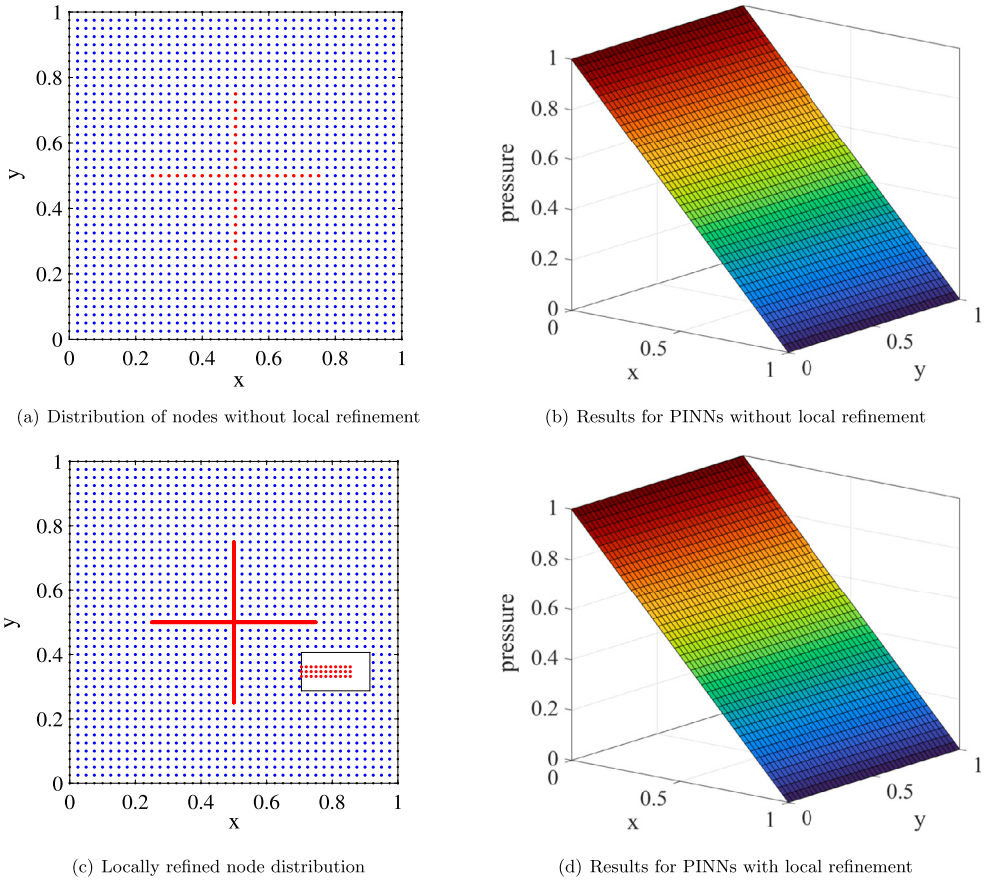


Fig. 11. Nodal distribution and results of PINNs.

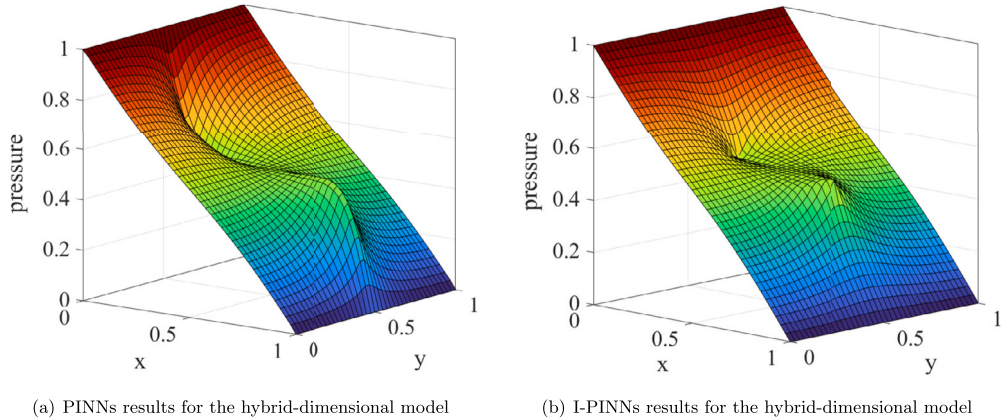


Fig. 12. Comparison of the results of the I-PINNs and the PINNs.

We use uniform rectangular grids ($N_x = N_y = 50$) to discretize the computational domain and make 800,000 iteration steps. The reference solution in Fig. 14(b) is in better agreement with the result we obtained in Fig. 15(a). Next, the variation of the MAE training loss for all epochs is plotted in Fig. 15(b), showing that the lowest loss is about 4.589×10^{-4} . Further, Fig. 16 quantifies the pressure at $y = 0.5$ and $x = 0.3$ profiles. By comparing the results of the reference solution and the I-PINNs, we find that there is not much difference between them. The maximum errors between the I-PINNs and the reference solution are 5.316×10^{-2} and 3.599×10^{-2} on the two line segments, indicating that the I-PINNs can be well applied to curved fractures.

Example 4.4 (Regular fracture network). This example [45] simulates the regular fracture network in a square domain $[0, 1] \times [0, 1]$. The fractures are located at $x = 0.5$, $x = 0.625$, $x = 0.75$, $y = 0.5$, $y = 0.625$ and $y = 0.75$, and each fracture has thickness $\epsilon = 10^{-4}$. The

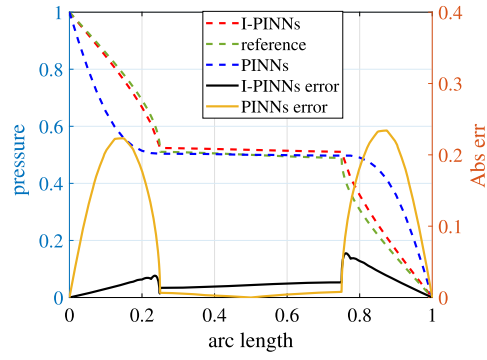
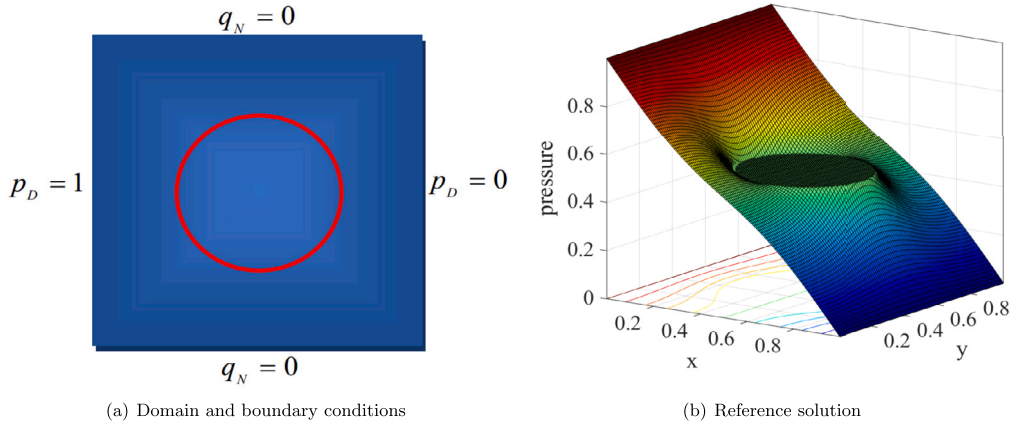
Fig. 13. Comparison of results using various methods on the line $y = 0.5$.

Fig. 14. Boundary conditions and reference solution.

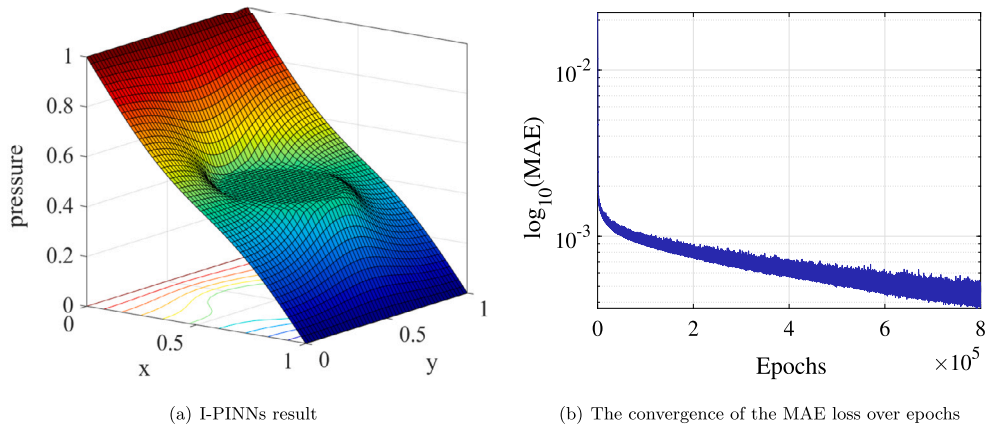


Fig. 15. I-PINNs result and convergence curve of the loss function.

permeabilities of matrix and fracture are 1 and 10^4 , respectively. In addition, the left boundary is the inflow boundary with $q_N = 1$, while the right boundary is the Dirichlet boundary with $p_D = 1$. The other boundaries are set to be impermeable, i.e., $q_N = 0$ (as shown in Fig. 17(a)).

To verify the accuracy of I-PINNs in the case of multiple complex fracture networks. We use 2500 uniform rectangular meshes to discretize the computational domain and make half a million iteration steps. The reference solution given in Fig. 17(b) matches well with the numerical pressure in Fig. 18(a). In addition, we find from Fig. 18(b) that it is very difficult to accurately describe the pressure of each fracture when simulating such complex fractures with conventional PINNs, thus also indicating that our proposed model is quite advantageous in simulating complex fracture networks. The variation of the MAE training loss for all epochs is plotted

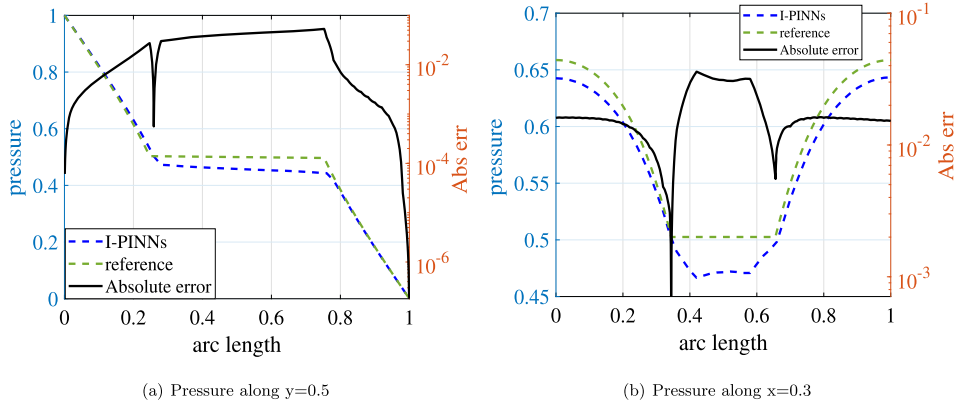


Fig. 16. Pressure profiles along different lines.

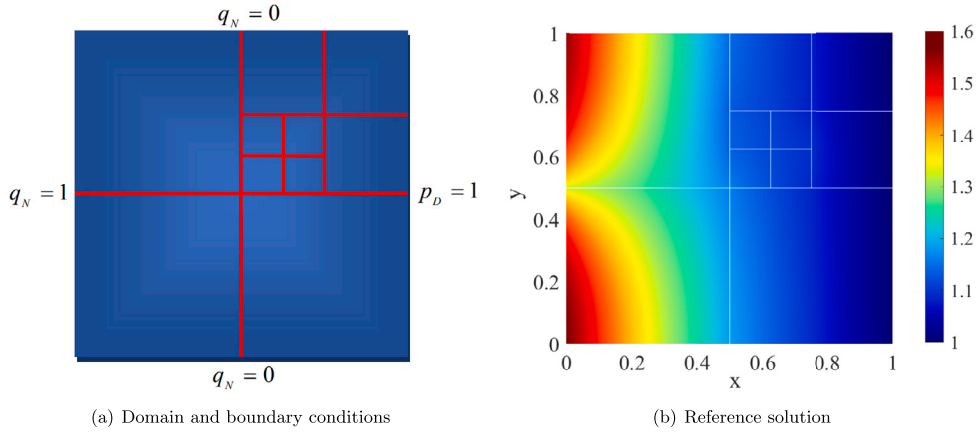


Fig. 17. Boundary conditions and reference solution.

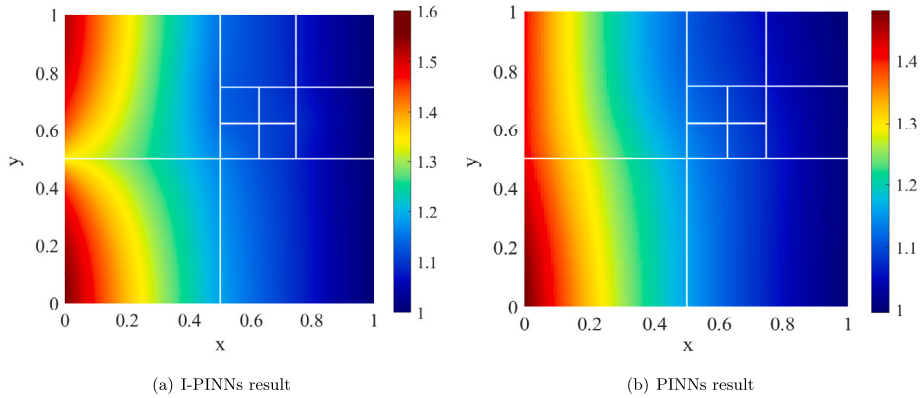


Fig. 18. Comparison of results for PINNs and I-PINNs.

in Fig. 19, showing that the lowest loss is about 1.944×10^{-4} . Similar to the previous examples, Fig. 20 shows the pressure along $y=0.7$ and $x=0.5$. By comparing the results of the reference solution and the I-PINNs, it is found that the two curves are very close to each other. The maximum errors between the I-PINNs and the reference solution are 1.70×10^{-2} and 2.5×10^{-3} on the two line segments. The results show that for complex multiple fracture models, the I-PINNs also have good accuracy.

Example 4.5 (A realistic case). This example is a benchmark problem in [45], modified from a real set of fractures from an interpreted outcrop in the Sotra island. It is a complex fracture network containing 63 fractures with different lengths and connectivity. Following the set up in [19], the computational region was uniformly shrunk to $\Omega = [0, 1.4] \times [0, 1.2]$, the top and bottom boundaries are

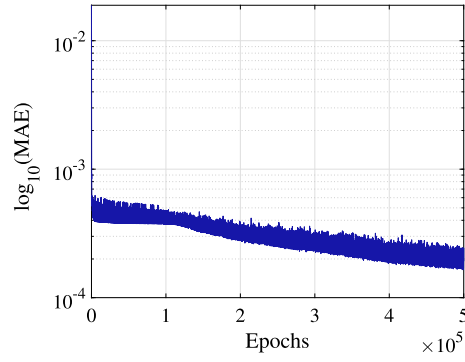


Fig. 19. The convergence of the MAE loss over epochs.

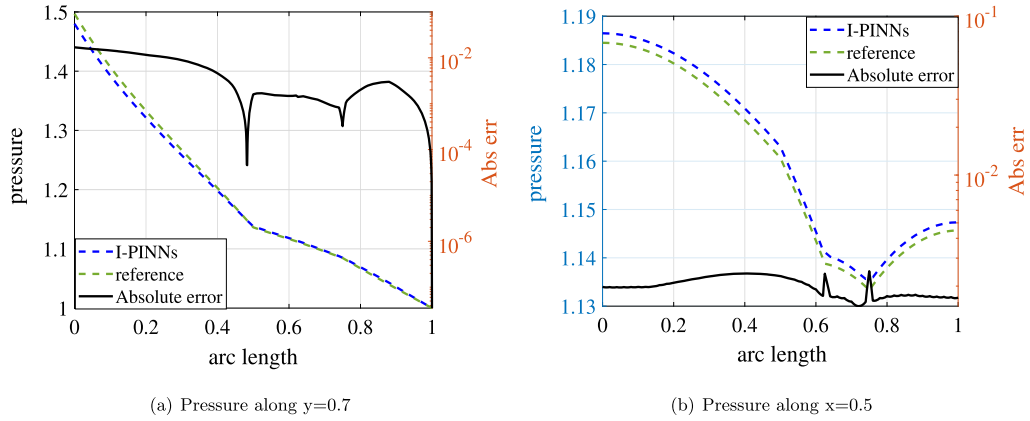


Fig. 20. Pressure profiles along different lines.

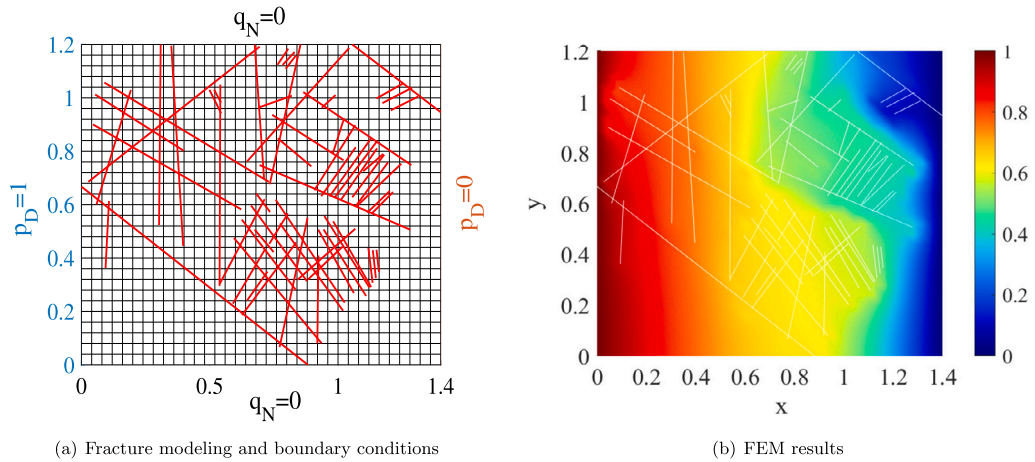


Fig. 21. Fracture modeling, boundary conditions and FEM results.

impermeable, i.e. $q_N = 0$, while the left and right boundaries are Dirichlet ones with $p_D = 1$ and $p_D = 0$, respectively. The regional settings and the fractures on it are presented in Fig. 21(a). These fractures have uniform thickness of $\epsilon = 2.5 \times 10^{-4}$ and permeability of 10^4 , while the matrix permeability is 1. Detailed geometric data for the fractures can be found in Ref. [18], while we just reduce their geometry uniformly.

In order to verify that the I-PINNs proposed in this paper can simulate real fracture cases, we will arrange 35×30 uniform rectangular mesh division areas. In I-PINNs, we use a 4-layer fully connected network with 600 neurons per hidden layer and iterate one million steps. It's great to see that the results of FEM and I-PINNs are in good agreement in Figs. 21(b) and 22(a). In both cases,

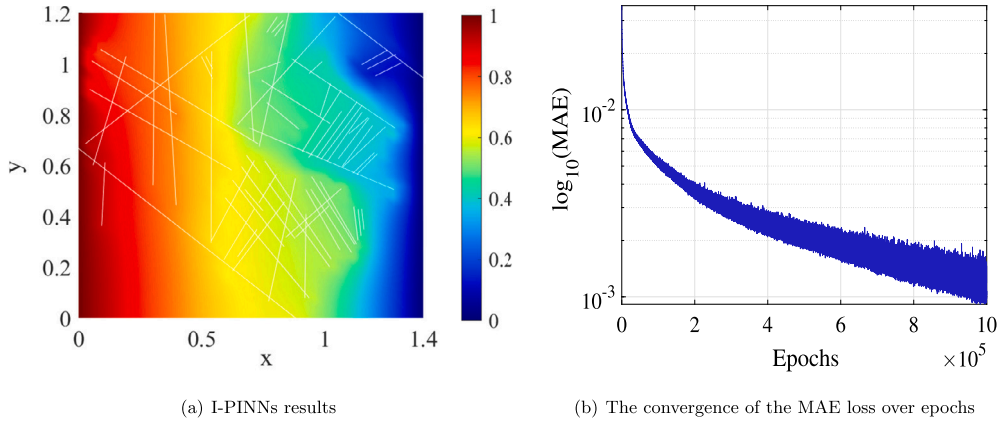


Fig. 22. I-PINNs results and loss function convergence curve.

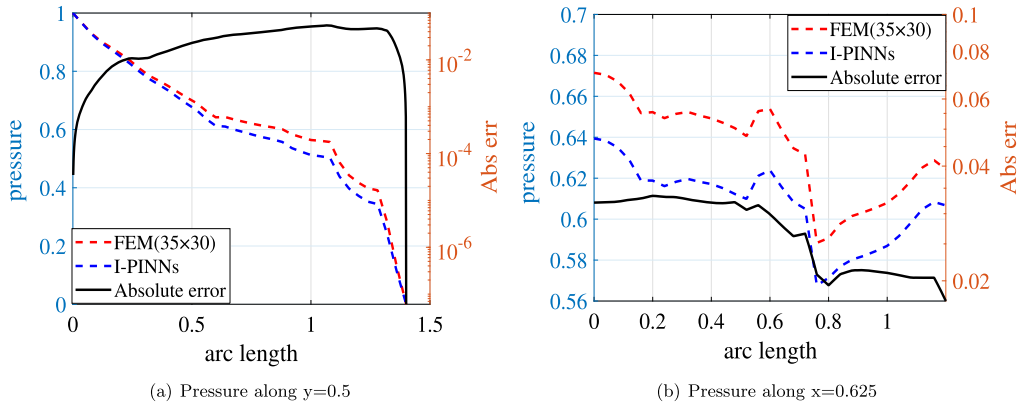


Fig. 23. Pressure profiles along different lines.

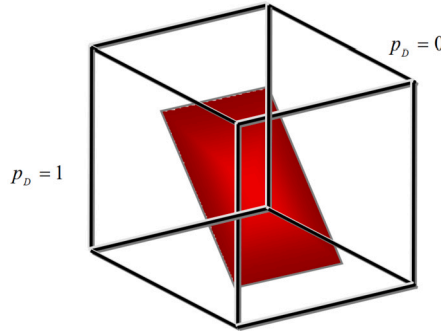


Fig. 24. 3D Domain and boundary conditions.

the same mesh was used for analysis, which further strengthens the comparison between the two methods. This agreement suggests that I-PINNs can simulate the real fracture situation very well. As usual, we still give the loss convergence curve for this problem (as in Fig. 22(b)). In addition, Fig. 23 shows the pressure along $y = 0.5$ and $x = 0.625$. By comparing the finite element results and the I-PINNs, we find that the two curves are very close. The maximum errors between the I-PINNs and the FEM results are 5.453×10^{-2} and 3.344×10^{-2} on the two line segments. Through the above analysis we can conclude that I-PINNs still have good accuracy, convergence performance in simulating complex real fracture problems.

Example 4.6 (3D case). In the last example, we simulate a single plane fracture in the 3D cubic domain $[0, 1] \times [0, 1] \times [0, 1]$, and the four vertex coordinates of the fracture are $(0.25, 0.2, 0.1)$, $(0.25, 0.7, 0.7)$, $(0.75, 0.7, 0.7)$ and $(0.75, 0.2, 0.1)$. The permeability of matrix is 1, and the thickness and permeability of fracture are 10^{-3} and 10^4 , respectively. In addition, the left and right boundaries of the calculation domain meet the Dirichlet boundary conditions $p_D = 1$ and $p_D = 0$, respectively. The other boundaries are impermeable (see Fig. 24).

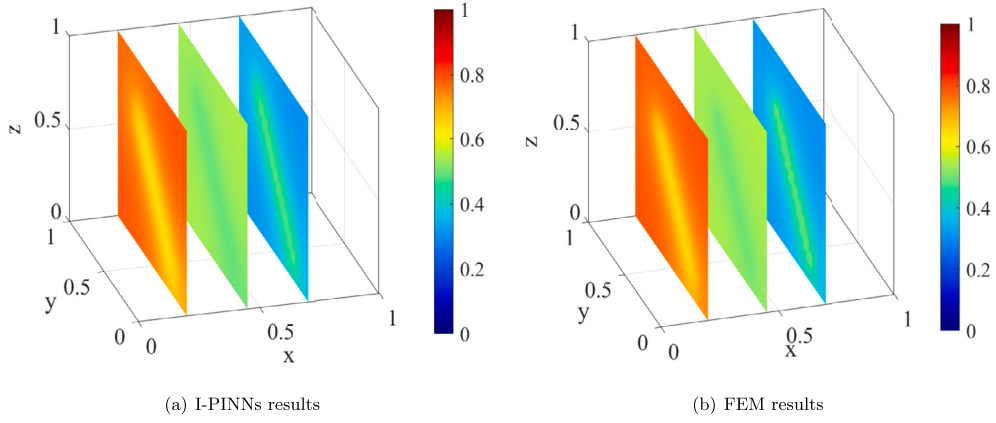


Fig. 25. Slice planes in x-direction of pressure in 3D space.

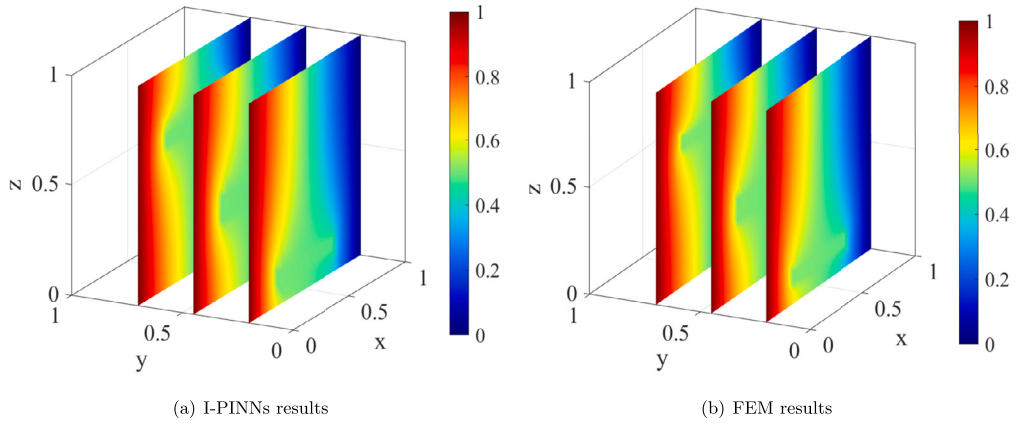


Fig. 26. Slice planes in y-direction of pressure in 3D space.

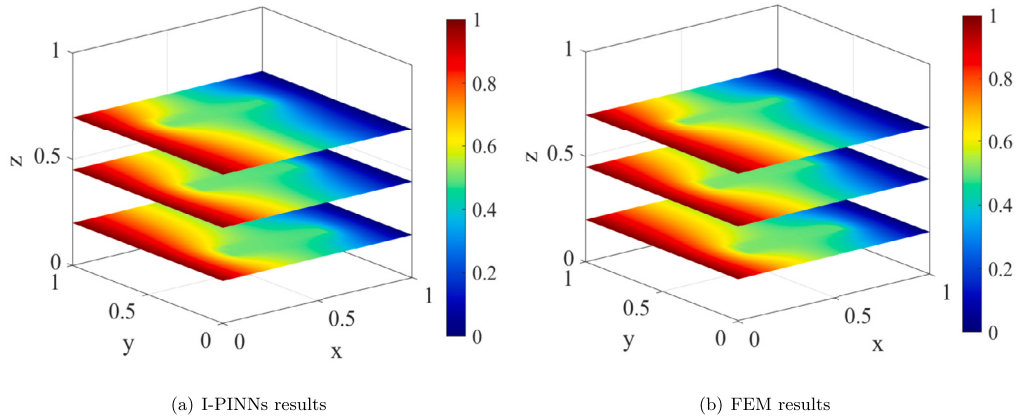


Fig. 27. Slice planes in z-direction of pressure in 3D space.

To verify that training the 3D model with the I-PINNs architecture still yields good target results. The cubic structure was divided into $20 \times 20 \times 20$ meshes. Then we constructed 2 hidden layers with 300 neurons in each hidden layer. The total iteration step is set to be 100,000. It should be noted that the hybrid-dimensional representation of the permeability tensor in the 3D model from [18] is $\mathbf{K} = \mathbf{K}_m + \sum_{i=1}^L \epsilon_i k_{f_i} \delta_i(\cdot) \mathbb{1}_i(\cdot) (\mathbf{I} - \sigma_i \sigma_i^T)$. Figs. 25–27 illustrate the slice planes in the coordinate axes-direction of pressure in 3D space. From the figures we can see that there are essentially the same between the finite element results on the coarse mesh and I-PINNs results, indicating that the I-PINNs can be successfully extended to 3D problems.

Algorithm 1 Bayesian personalized ranking based latent feature embedding model.**Input:** simulation time T , time step Δt , training tolerance ε , and maximum epoch number M **Output:** c^*

```

1: while  $t < T$  do
2:   if  $t = \text{first time}$ : initialize NN by using Xavier method, and input IC
3:   if  $t > \text{first time}$ : initialize NN by using the trained NN of previous time step
4:   update equation parameters
5:    $m = 1$ 
6:   while  $m < M$  and  $\text{MAE} > \varepsilon$  do
7:     input pressure at previous time step  $c^t$  into NN to get current approximate pressure  $c^*$ 
8:     evaluate loss MAE by using Eq. (5.7)
9:     optimize NN parameters according to MAE
10:     $m = m + 1$ 
11:   end while
12:   take  $c^*$  associated with the lowest loss as  $c^{t+1}$ 
13:    $t = t + \Delta t$ 
14: end while

```

5. Applications in contaminant transportation in fractured porous media

In this section, we continue to explore in depth the contaminant transport in fractured porous media with the novel discrete fracture model. The governing equations of the contaminant transportation reads

$$\phi c_t + \nabla \cdot (\mathbf{u}c) - \nabla \cdot (\mathbf{D}(\mathbf{u})\nabla c) = \tilde{c}q, \quad (x, t) \in \Omega \times (0, T], \quad (5.1)$$

$$(\mathbf{u}c - \mathbf{D}(\mathbf{u})\nabla c) \cdot \mathbf{n} = c_{in} \mathbf{u} \cdot \mathbf{n}, \quad (x, t) \in \Gamma_{in} \times (0, T], \quad (5.2)$$

$$(-\mathbf{D}(\mathbf{u})\nabla c) \cdot \mathbf{n} = 0, \quad (x, t) \in \Gamma_{out} \times (0, T], \quad (5.3)$$

$$c(x, 0) = c_0(x), \quad x \in \Omega, \quad (5.4)$$

where c is the concentration of the contaminant, ϕ is the porosity of the porous media, $\mathbf{u}$ is the Darcy's velocity of the flow, $\mathbf{D}(\mathbf{u})$ is the diffusion coefficient of the contaminant, q is the source term, $\tilde{c} = c$ if $q < 0$ and $\tilde{c} = c_{inject}$ is the inflow concentration from the source term if $q > 0$, Γ_{in} is the inflow boundary and Γ_{out} is the outflow boundary, c_{in} is the concentration of the contaminant on the inflow boundary, c_0 is the initial concentration of the contaminant. The time derivative is discretized by the forward Euler method. In addition, for the numerical solutions of concentrations beyond the range of $[0, 1]$, we use the I-PINNs bounds-preserving method for the treatment. This method is based on the idea of data normalization, which maps the concentration c to the $[0, 1]$ range. The normalization equations and examples are shown below:

$$c = \frac{c - c_{min}}{c_{max} - c_{min}}. \quad (5.5)$$

The fully-discrete FEM formulation is to find $c \in H_0^1$, s.t. $\forall \eta \in H_0^1$

$$\int_{\Omega} \phi \frac{c^{t+1} - c^t}{\Delta t} \eta dx dy - \int_{\Omega} (\mathbf{u}c^{t+1}) \cdot \nabla \eta dx dy + \int_{\Omega} (\mathbf{D}(\mathbf{u})\nabla c^{t+1}) \cdot \nabla \eta dx dy = \int_{\Omega} \tilde{c}q \eta dx dy - \int_{\Gamma_{in}} c_{in} \mathbf{u} \cdot \mathbf{n} \eta ds. \quad (5.6)$$

Furthermore, the output of the NN is substituted into Eq. (5.6) to approximate the PDE residual which should be minimized. Therefore, mean absolute error (MAE) loss function for the NN is defined as

$$\begin{aligned} Loss_f = \text{torch.abs} \left(\int_{\Omega} \phi \frac{c^* - c^t}{\Delta t} \eta dx dy - \int_{\Omega} (\mathbf{u}c^*) \cdot \nabla \eta dx dy + \int_{\Omega} (\mathbf{D}(\mathbf{u})\nabla c^*) \cdot \nabla \eta dx dy - \int_{\Omega} \tilde{c}q \eta dx dy \right. \\ \left. + \int_{\Gamma_{in}} c_{in} \mathbf{u} \cdot \mathbf{n} \eta ds \right). \text{mean}(), \end{aligned} \quad (5.7)$$

where c^* is the NN output that represents c^{t+1} . It is worth noting that we need to use the NN results trained in the previous time step as the input values for the initial NN in the current time step. The complete training algorithm is as follows (Algorithm 1).

5.1. The bound-preserving test

In this example, we simulate the transport of contaminants in the porous media containing fracture and verify the significance of the bound-preserving technique in simulating miscible displacements in porous media. Let $\Omega = [0, 2\pi] \times [0, 2\pi]$. The initial concentrations c_0 is

$$c_0 = \begin{cases} 1, & 0 \leq x \leq \frac{\pi}{2}, \frac{3\pi}{4} \leq y \leq \frac{5\pi}{4}, \\ 0, & \text{otherwise.} \end{cases}$$

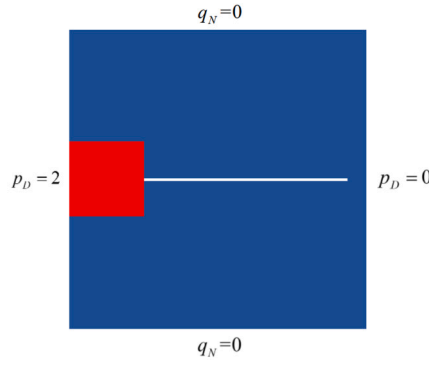


Fig. 28. Domain and boundary conditions.

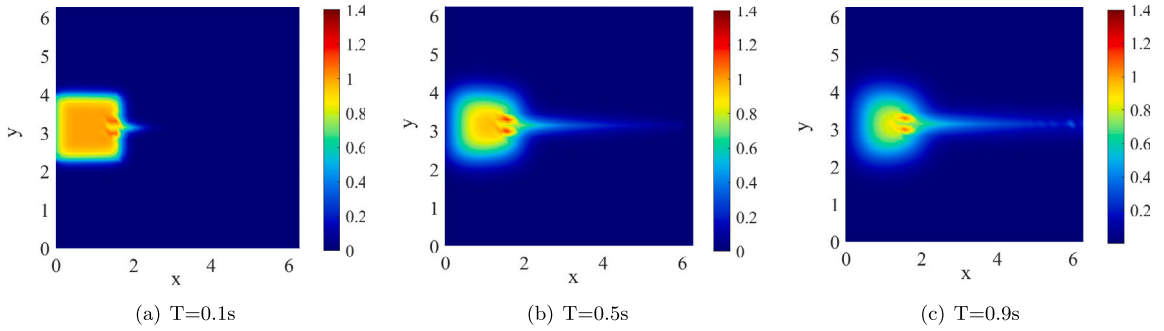


Fig. 29. Concentration c without boundary-preserving.

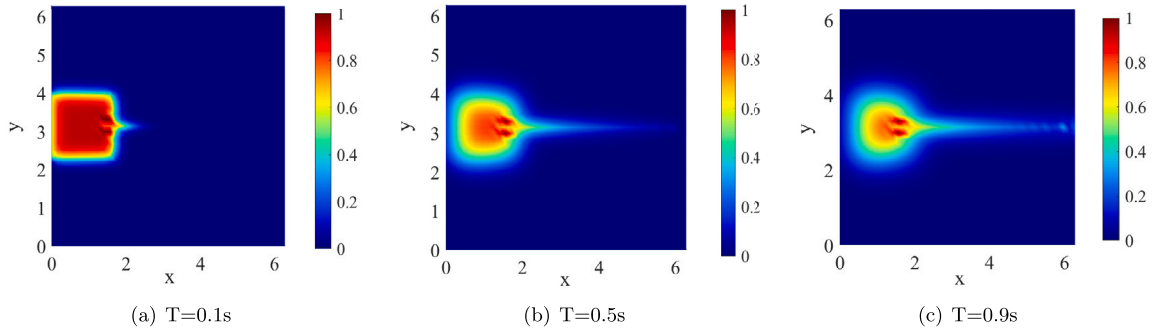


Fig. 30. Concentration c with boundary-preserving.

The left and right endpoints of the fracture are $(\frac{\pi}{2}, \pi)$, $(\frac{15\pi}{8}, \pi)$, respectively. Other parameters are taken to be $\phi = 1$, $\mathbf{K}_m = \mathbf{I}$, $\epsilon k_f = 10^2$, $\mathbf{D} = 0.1\mathbf{I}$, $c_{in} = 0$. Moreover, for pressure equation, the left and right boundaries are Dirichlet boundaries with pressure $p_{Dleft} = 2$, $p_{Dright} = 0$, while the bottom and top boundaries are Neumann boundaries with $p_N = 0$. The left boundary is the inflow boundary with $c_{in} = 0$. The right boundary is the outflow boundary. For a detailed description, see Fig. 28.

In this example, we calculate the numerical results for $T = 0.1$ s, $T = 0.5$ s, and $T = 0.9$ s. It should be noted that we divided the computational domain into 40×40 uniform rectangular meshes and the time interval is set to $dt = 0.1$ s. For I-PINNs, we used 2 hidden layers with 200 neurons per layer and 200,000 iteration steps. To test the effectiveness of the boundary-preserving technique, we simulated the results without the I-PINNs boundary-preserving technique (shown in Fig. 29), and the numerical results after adopting it (shown in Fig. 30). We can conclude that the boundary-preserving technique is effective and that the concentration c after bound preservation is between 0 and 1.

6. Conclusion

In this paper, we proposed the novel I-PINNs framework and applied it to RDFM for single-phase flow. Meanwhile, we proposed a boundary-preserving technique based on I-PINNs and successfully applied it to simulate the contaminant transportation in fractured porous media. Compared with the traditional PINNs, I-PINNs could get rid of the large errors in the computation of the extreme

heterogeneous problems using the automatic differentiation technique, and at the same time, avoided the process of matrix inversion compared to the traditional numerical methods. Additionally, it had good agent modeling characteristics.

The examples presented in this paper demonstrated that the method based on I-PINNs was able to solve the fracture problem efficiently and with very high accuracy. It was also important to emphasize that the I-PINNs-based approach could easily be used to deal with complex fracture networks and 3D fracture structures since the variety of training parameters could be kept almost constant as the number and dimension of the fractures increased. In order to gain insight into the model, we also analyzed and quantified the performance of the I-PINNs model by selecting different underlying network architectures. Nevertheless, I-PINNs still have some limitations in modeling fractured porous media problems. For example, it was not computationally efficient when dealing with multiple tiny fractures. Currently, we have incorporated GPU parallelism into our approach to solve this problem. In conclusion, the developed I-PINNs framework could be considered as a powerful alternative to the single-phase flow problem in fractured porous media.

CRediT authorship contribution statement

Chao Wang: Writing – original draft, Software, Methodology. **Hui Guo:** Writing – review & editing, Supervision, Investigation, Funding acquisition. **Xia Yan:** Methodology. **Zhang-Lei Shi:** Methodology. **Yang Yang:** Writing – review & editing, Validation, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Hui Guo reports financial support, equipment, drugs, or supplies, and travel were provided by National key Nature Science Foundation of China. Hui Guo reports financial support, equipment, drugs, or supplies, and travel were provided by Natural Science Foundation of Shandong Province. Yang Yang reports travel was provided by Simons Foundation. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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